SIYI WANG

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EDUCATION BACKGROUND

RICE UNIVERSITY, DEPARTMENT OF ECONOMICS ***HOUSTON, US***
Ph.D. Candidate in Economics *Aug 2021 - Present*

- **Relevant Coursework:** Computational Economics, Empirical Microeconomics, Topics in Economic Theory I, Financial Economics I, Economics of Insurance, Economic Growth & Development, Matching and Market Design (auditing), Microeconomics I, Microeconomics II, Econometrics I, Econometrics II, Macroeconomics, Introduction to Analysis

TSINGHUA UNIVERSITY (THU), SCHOOL OF ECONOMICS AND MANAGEMENT (SEM) ***BEIJING, CHINA***
Bachelor in Economics, Major in Economics and Finance *Sep 2017 - Jun 2021*

UNIVERSITY OF PENNSYLVANIA (UPENN), DEPARTMENT OF ECONOMICS ***PHILADELPHIA, US***
Exchange Student *Aug 2019 - July 2020*

RESEARCH INTERESTS

Applied Microeconomics: Health Economics, Industrial Organization

PUBLICATION

ZHANG, Z., LI, K., WANG, S., FASHAW-WALTERS, S., & HOU, Y. (2024, NOVEMBER). CHANGE OF OWNERSHIP AND QUALITY OF HOME HEALTH AGENCY CARE. IN JAMA HEALTH FORUM (VOL. 5, NO. 11, PP. E243767-E243767). AMERICAN MEDICAL ASSOCIATION.

RESEARCH PROJECTS

ESTIMATING TRANSFERABLE UTILITY MATCHING GAMES WITH A NESTED FIXED POINT APPROACH USING JULIA ***HOUSTON, US***
Coding project for the course "Computational Economics" (Score: 9/10) *Aug 2022 - Sep 2022*

- Built from scratch a transferable utility model of one-to-one, two-sided matching with an equal and finite number of agents on both sides of the market (the econometric part of the model is based on Fox, Yang and Hsu (2018))
- Simulated dataset with true model parameters, wrote the linear programming code to solve for the equilibrium matches, and computed the 87 moments and the GMM objective function
- Used a global optimization solver to optimize the GMM objective function embedded with the nested fixed-point problem to estimate the parameters of the model
- Carried out Monte Carlo study to compute bias, RMSE, and confidence interval coverage of the estimated results
- Utilized parallelization on the cluster to reduce the computational burden

ESTIMATING BLP DEMAND USING JULIA ***HOUSTON, US***
Coding project for the course "Computational Economics" (Score: 7/10) *Oct 2022 - Nov 2022*

- Estimated the demand for differentiated products using the mathematical program with equilibrium constraints (MPEC) approach (the baseline model is based on Berry, Levinsohn and Pakes (1995))
- Generated a fake dataset with T = 10 markets and J = 10 inside goods per market and solved a system of demand and pricing equations to compute a Nash-Bertrand equilibrium
- Estimated the means and variances of the random coefficients using the MPEC formulation the problem definition language JuMP
- Computed a merger counterfactual describing what happened if inside goods 1 and 2 merged and the merged firm chose prices for both goods 1 and 2

CALIBRATING REAL BUSINESS CYCLE MODEL WITH POST-WAR U.S. ECONOMY STATISTICS USING MATLAB ***PHILADELPHIA, US***
Final project for the course "Numerical Methods for Macroeconomists" *Mar 2020 - Apr 2020*

- Set up a real business cycle model (based on Kydland and Prescott (1982) & Greenwood, Hercowitz and Huffman (1988) and solve for the solution using discrete state space dynamic programming method with ***MATLAB***
- Calibrated the model by matching model-generated business cycle statistics with post-war U.S. economy statistics

ESTIMATING THE PERFORMANCE OF TELE-SECONDARY SCHOOL AND NORMAL SCHOOL IN MEXICO WITH GENERALIZED ROY MODEL USING R ***PHILADELPHIA, US***
Final paper for the course "Microeconometrics" *Apr 2020 - May 2020*

- Formed an empirical essay comparing the effectiveness of tele-secondary education (delivering academic materials online) with regular schooling in Mexico using a Roy model framework to mitigate selection bias

- Built the model with R from constructing a Probit discrete choice model for types of schools chosen, forming the control function, estimating counterfactuals, to assessing TT, TUT and ATE
- Improved the traditional baseline model by adding a new interaction term of distance difference times income in probit model, which showed a significantly better AUC performance in classification

USING MULTIPLE MACHINE LEARNING METHODS TO ESTIMATE INTERGENERATION INCOME MOBILITY OF U.S. CENSUS TRACTS WITH "OPPORTUNITY ATLAS" DATABASE USING R ***PHILADELPHIA, US***

Independent final project for the course "Modern Data Mining" *Nov 2019 - Jan 2020*

- Built and compared among 3 classification models (multiple logistic regression using LASSO/exhaustive search, and random forest) to identify which type of census tracts promotes upward income mobility
- Predicted upward and downward intergeneration mobility of different races in all census tracts using linear regression with LASSO, single tree, and random forest, and selected the model with the best test performance

RESEARCH ASSISTANT EXPERIENCES

"INFORMATIVE" VS. "PERSUASIVE"? EXAMINING INDUSTRY PAYMENTS TO PHYSICIANS WITH OPEN PAYMENT DATABASE ***PHILADELPHIA, US***

Supervisor: Hanming Fang, Joseph Cohen Term Professor, UPenn Economics Department *Apr 2020 – Sep 2020*

- Designed a time- and space-efficient method to distinguish two hypotheses, by grouping millions of payments observations into thousand therapeutic-similar clusters by wrangling 5 related databases
- Excogitated a new mechanism to tell apart two competing hypotheses leveraging variance in dispensing restrictions in pharmaceutical industry, which was inspired by related chapters in *Handbook of Health Economics*
- Working on machine learning methods to cluster physicians and drugs in a more accurate way

CHINESE FINANCIAL PROPAGATION ***BEIJING, CHINA***

Supervisor: Tuo Chen, Assistant Professor, SEM, THU *Jun 2019 - Aug 2019*

- Constructed from scratch a financing database of Chinese A-share listed enterprises with 100,000+ firm-year observations including categories of corporate bank loans, bond issuances, shadow banking loans, related party transactions, etc.
- Cleaned collected data, carried out fundamental EDA, and performed linear regression analysis and model selection with R
- Completed a review of 30+ journal papers on financial propagation and communicated the summary of findings to the professor

CODING SKILLS

- Language: high proficiency in Julia, R, MATLAB and basic familiarity with C++
- Experienced with optimization algorithms and implementations: linear programming, nonlinear programming, and global optimization
- Experienced with machine learning algorithms and implementations: neural network, tree-based methods, support vector machine, and random forest.
- Additional coding experience with text mining sentiment analysis
- Abundant experience with data collecting, data cleaning and EDA